

Constraint-Driven Closure and the Yang–Mills Mass Gap

Paper 6: capstone bridge note on admissibility and non-primitive demands

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February 26, 2026 (Paper 6, R5)

Abstract

Route 1 constructs, at fixed positive resolution $\ell > 0$, an Osterwalder–Schrader (OS) positive Euclidean theory on a bounded local net of gauge-invariant observables and proves exponential clustering and a Hamiltonian spectral gap in the OS reconstruction, conditional on explicit ultraviolet (UV) identification inputs, including existence of a macroscopic scaling limit along a tuned UV trajectory [1]. Papers 2–5 isolate the interfaces supporting this conclusion [2, 3, 4, 5]: finite-depth entrance into a uniform contraction basin supplies macroscopic clustering without stable-manifold renormalization-group control; the spectral gap is extracted from OS semigroup time; the UV identification assumptions are orthogonal to the entrance mechanism [4]; and pointlike local fields correspond to a strict ultraviolet scope enlargement beyond the bounded macroscopic net.

This note makes explicit that any gap statement is relative to (i) a declared observable scope and (ii) a declared invariant package. We define an admissibility datum encoding these declarations and record an elementary closure observation: if a further demand is not invariant under the equivalence induced by the declared invariant package, then it cannot be treated as a same-scope consequence; imposing it amounts to enlarging the scope and/or the invariant package. Applied to the Route 1 program at macroscopic scope Σ_ℓ , this classifies stable-manifold RG structure and sharp $t \rightarrow 0$ local fields as optional upgrades beyond the primitive data required for the Hamiltonian gap statement on $\mathcal{O}_\ell$. Under an admissibility datum that treats Σ_ℓ with invariants (M1)–(M4) as primitive, Route 1 together with the UV package of Paper 4 supplies the required macroscopic invariant package.

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1 Purpose and placement in the arc

The Route 1 technical paper is written entirely in standard mathematical physics language and is intended to stand on its own. It constructs a continuum macroscopic Euclidean theory at fixed resolution $\ell > 0$ on a bounded net of gauge-invariant observables and proves a positive Hamiltonian spectral gap in OS reconstruction, conditional on explicit UV identification inputs.

The role of the present capstone is not to modify or extend Route 1 estimates. Its purpose is to make explicit a further ingredient that is often left implicit in discussions of “what counts” as a solution: the *problem statement* depends on a declared observable scope and a declared invariant package.

Remark 1.1 (Separation discipline). This capstone is part of Track B (bridge). It does not introduce any bridge terminology into the Route 1 technical argument. It is written to be logically downstream of Route 1 and Papers 2–5.

Interfaces already isolated in Papers 2–5

We treat the following as established at the level of structural interfaces:

- (I1) **Entrance versus stable manifolds (Paper 2):** finite-depth entrance into a uniform Dobrushin contraction basin is a sufficient control primitive for macroscopic exponential clustering; stable-manifold RG control is stronger and not logically required for the macroscopic clustering conclusion.
- (I2) **Semigroup spectral extraction (Paper 3):** OS positivity and exponential decay of Euclidean time-separated correlations on a bounded net imply a Hamiltonian spectral gap for the OS-reconstructed time-translation generator.
- (I3) **UV identification package (Paper 4):** the interpretation of the macroscopic statement in physical units requires explicit UV assumptions; these assumptions are orthogonal to the entrance mechanism.
- (I4) **Sharp fields as scope enlargement (Paper 5):** pointlike local fields (sharp $t \rightarrow 0$ limits) require additional ultraviolet regularity inputs and are not forced by the macroscopic Hamiltonian gap statement on $\mathcal{O}_\ell$.

The present capstone adds a scope bookkeeping layer and uses it to classify certain common additional demands.

Item	Where stated	Status	Function
UV trajectory existence and macroscopic scaling limit at fixed ℓ	Route 1 [1], Ass. 1(U1); Paper 4 [4], Ass. 3.1(U2)	External input	UV Provides a macroscopic limit functional $\mathcal{S}$ on $\mathcal{A}_\ell$.
Non-collapse of the physical decay rate / mass scale	Route 1 [1], Ass. 1(U3); Paper 4 [4], Ass. 3.1(U4)	External input	UV Prevents collapse of the physical-time clustering rate as $a \downarrow 0$.
Reference contractivity $\delta(\Gamma^{\text{ref}}) \leq q_* < 1$ for the α -tilted reference specification	Route 1 [1], Ass. 2(R1)	Internal module assumption	as- Supplies a uniform contraction basin for the transfer gate.
Cut-local surrogate / transfer framework for the effective tilt	Route 1 [1], Ass. 3	Internal module assumption	as- Controls the exact-to-reference defect and yields the χ^2 gate.
OS reconstruction regularity (bounded-net version of OS5)	Paper 4 [4], Ass. 3.1(U5); Paper 3 [3]	Standard reconstruction package	re- Ensures a strongly continuous contraction semigroup e^{-tH} .

Table 1: Status map of the principal nontrivial inputs. “External UV input” means not derived by the finite-depth entrance mechanism; “internal module assumption” means a model-dependent comparison component stated explicitly as a hypothesis in Route 1.

Status of key inputs across the arc

The six-paper arc separates (i) *proved* contraction/semigroup mechanisms from (ii) *internal comparison modules* and (iii) *external UV identification inputs*. Table 1 records the status of the main nontrivial inputs used to obtain the macroscopic invariant package (M1)–(M4).

Remark 1.2 (Dominant external input). The most significant external input in the arc is the existence of a tuned UV trajectory with a macroscopic scaling limit at fixed $\ell > 0$ (first row of Table 1). This is intentionally isolated and not addressed by the finite-depth entrance mechanism itself.

2 Declared scope and invariant package

2.1 Macroscopic scope at fixed resolution

Definition 2.1 (Macroscopic scope Σ_ℓ). Fix $\ell > 0$. The *macroscopic scope* Σ_ℓ is the bounded local net

$$\Lambda \mapsto \mathcal{O}_\ell(\Lambda)$$

of gauge-invariant bounded observables at resolution ℓ , with inductive-limit $*$ -algebra

$$\mathcal{A}_\ell := \bigcup_{\Lambda} \mathcal{O}_\ell(\Lambda).$$

As part of the declared scope we fix a translation action $(\tau_x)_{x \in \mathbb{R}^4}$ by $*$ -automorphisms of $\mathcal{A}_\ell$ and a time reflection $*$ -automorphism ι_θ . We set $\Theta(F) := \iota_\theta(F^*)$ and write $\mathcal{A}_{+,\ell}$ for the associated positive-time subalgebra used in OS reconstruction.

Remark 2.2. At this scope, all probes are bounded and have an a priori ultraviolet cutoff at resolution ℓ . In particular, no pointlike fields or unbounded operator-valued distributions are part of the scope.

2.2 Invariant content at macroscopic scope

Definition 2.3 (Macroscopic invariant package). A realization at scope Σ_ℓ satisfies the *macroscopic invariant package* if it provides:

- (M1) an OS-positive Euclidean functional $\mathcal{S}$ on $\mathcal{A}_\ell$ with respect to the scope structures (τ_x, Θ) ;
- (M2) exponential Euclidean time-clustering for bounded observables in $\mathcal{A}_{+, \ell}$;
- (M3) an OS reconstruction $(\mathcal{H}, \Omega, H)$ with $H \geq 0$ self-adjoint (cf. [6, 13]);
- (M4) a Hamiltonian spectral gap: $\text{Spec}(H) \subset \{0\} \cup [m_*, \infty)$ for some $m_* > 0$.

Remark 2.4 (Terminology). Throughout this bridge arc, “mass gap” is used in the sense of (M4): a Hamiltonian spectral gap above the vacuum in the representation generated by $\mathcal{A}_\ell$. No claim is made here about the full relativistic energy–momentum spectrum beyond what is encoded by OS reconstruction at fixed macroscopic scope.

3 Finite-depth entrance and majorant mixing as contraction witnesses

This section records how the Route 1 control primitive is expressed in bridge language.

3.1 Stable-manifold template versus entrance template

Remark 3.1 (Two templates). In the stable-manifold template, one studies an RG map $\mathcal{R}$ on a space of effective actions and controls a trajectory near a critical point via global dynamical objects (relevant/irrelevant splitting, stable manifolds, transport along all scales), cf. Wilson–Kogut [8] and Brydges [9]. In the entrance template, one proves a finite-depth statement: after a bounded number of blocking steps, the exact blocked specification lies in a fixed contraction basin (Dobrushin/Kotecký–Preiss) [7], after which standard contraction consequences apply.

Under the macroscopic invariants 2.3, the only relevant output of RG control is exponential clustering. The entrance estimate is therefore a natural contraction witness for (M2).

3.2 Majorant mixing: an intrinsic witness on compact groups

Route 1 uses an auxiliary compact-group semigroup (heat kernel) to quantify a mixing depth independent of the microscopic measure.

Definition 3.2 (Majorant mixing witness). Let G be a compact connected simple Lie group with Haar probability measure dg . Let $\{K_t\}_{t>0}$ be the heat-kernel semigroup on G . The *majorant mixing witness* at tolerance $\delta > 0$ is the unique time $s_\delta > 0$ such that

$$D_{\chi^2}(K_{s_\delta} \parallel dg) = \|K_{s_\delta} - 1\|_{L^2(dg)}^2 = \delta.$$

Remark 3.3 (Explicit spectral witness). Let $C_2(R)$ denote the quadratic Casimir eigenvalue in the irreducible unitary representation R of G and let d_R be its dimension. The heat kernel admits the character expansion

$$K_t(g) = \sum_R d_R \chi_R(g) e^{-tC_2(R)},$$

so that $K_t - 1 = \sum_{R \neq 1} d_R \chi_R e^{-tC_2(R)}$. By orthogonality of characters in $L^2(dg)$ one obtains

$$D_{\chi^2}(K_t \| dg) = \|K_t - 1\|_{L^2(dg)}^2 = \sum_{R \neq 1} d_R^2 e^{-2tC_2(R)}.$$

In particular, $t \mapsto D_{\chi^2}(K_t \| dg)$ is continuous and strictly decreasing with $D_{\chi^2}(K_t \| dg) \downarrow 0$ as $t \rightarrow \infty$ and $D_{\chi^2}(K_t \| dg) \uparrow \infty$ as $t \downarrow 0$ (since $\sum_{R \neq 1} d_R^2 = \infty$). Hence s_δ is well defined for every $\delta > 0$. See, e.g., Hall [10] for the expansion.

Remark 3.4 (Why “majorant” is not pointwise domination). In Route 1, the heat-kernel family provides an explicit contraction witness used to define a reference contraction basin. No pointwise domination of Wilson weights is asserted or used; the bridge to the exact blocked specification is supplied separately by a transfer gate (an L^2 or χ^2 budget).

4 Time without flow: OS semigroup spectral extraction

Paper 3 isolates the observation that a spectral gap is extracted from OS semigroup time, not from RG “time.”

Proposition 4.1 (Spectral extraction is semigroup-level). *(cf. Paper 3 [3]) Let $\mathcal{S}$ be an OS-positive Euclidean functional on $\mathcal{A}_\ell$ satisfying exponential Euclidean time-clustering on $\mathcal{A}_{+,\ell}$. Assume the OS reconstruction package holds so that $\mathcal{S}$ yields a strongly continuous contraction semigroup e^{-tH} on the reconstructed Hilbert space $(\mathcal{H}, \Omega)$. Then H has a spectral gap: there exists $m_* > 0$ such that*

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty).$$

Remark 4.2 (No renormalization “time” is required). The time parameter in Proposition 4.1 is the Euclidean time translation entering OS reconstruction. The proof uses only semigroup spectral theory plus clustering bounds. It does not require an RG flow coordinate or dynamical manifold structure.

5 Admissibility bookkeeping and scope enlargement

This section introduces a minimal bookkeeping device. It does not modify Route 1 estimates or alter the invariant package proved there. It records what it means, at a fixed observable scope, for an additional requirement to be logically independent of a declared invariant package.

5.1 Realizations, equivalence, and primitive demands

Definition 5.1 (Realization at macroscopic scope). A *realization* at macroscopic scope Σ_ℓ is a Euclidean functional $\mathcal{S}$ on $\mathcal{A}_\ell$ that is OS-positive with respect to the scope structures (τ_x, Θ) , i.e. reflection positivity holds on $\mathcal{A}_{+,\ell}$ for $\Theta(F) = \iota_\theta(F^*)$.

Definition 5.2 (Admissibility datum). An *admissibility datum* consists of:

- (A1) a declared scope Σ (here $\Sigma = \Sigma_\ell$);
- (A2) a declared invariant package $\mathcal{I}$ (here $\mathcal{I} = \{(\text{M1})-(\text{M4})\}$);
- (A3) an equivalence relation $\sim$ on realizations at scope Σ that identifies realizations which agree on the declared invariant package.

Example 5.3 (Canonical choice of $\sim$ at macroscopic scope). A canonical choice is: $X \sim Y$ if the Euclidean functionals agree on $\mathcal{A}_\ell$, i.e. $\mathcal{S}_X(F) = \mathcal{S}_Y(F)$ for all $F \in \mathcal{A}_\ell$. In this case the OS pre-Hilbert spaces are canonically isometric and the OS reconstructions are unitarily equivalent on the cyclic subspace generated by $\mathcal{A}_{+, \ell}$; see Lemma 5.4.

Lemma 5.4 (Equality of functionals implies unitary equivalence on the cyclic subspace). *Let X and Y be realizations at scope Σ_ℓ with $\mathcal{S}_X \equiv \mathcal{S}_Y$ on $\mathcal{A}_\ell$. Then the OS inner products coincide on $\mathcal{A}_{+, \ell}$:*

$$\langle F, G \rangle_{\text{OS}, X} = \mathcal{S}_X(\Theta F G) = \mathcal{S}_Y(\Theta F G) = \langle F, G \rangle_{\text{OS}, Y} \quad (F, G \in \mathcal{A}_{+, \ell}).$$

Consequently the null spaces coincide, the map $[F]_X \mapsto [F]_Y$ defines an isometry of the OS pre-Hilbert spaces, and it extends uniquely to a unitary between the completed OS Hilbert spaces on the cyclic subspaces generated by $\mathcal{A}_{+, \ell}$.

Proof. The displayed identity follows directly from the hypothesis $\mathcal{S}_X \equiv \mathcal{S}_Y$ on $\mathcal{A}_\ell$. In particular, $\langle F, F \rangle_{\text{OS}, X} = 0$ iff $\langle F, F \rangle_{\text{OS}, Y} = 0$, so the null spaces coincide and the assignment $[F]_X \mapsto [F]_Y$ is well defined and preserves the inner product. Extension to a unitary on the Hilbert space completions is standard. $\square$

Definition 5.5 (Primitive demand). A demand $\mathcal{D}$ on realizations at scope Σ is *primitive* (relative to the admissibility datum) if it is invariant under $\sim$: whenever $X \sim Y$, one has $\mathcal{D}(X) \iff \mathcal{D}(Y)$. Equivalently, $\mathcal{D}$ factors through the $\sim$ -equivalence class determined by the invariant package $\mathcal{I}$.

Definition 5.6 (Scope/invariant-package enlargement). A non-primitive demand at scope Σ can be incorporated only by enlarging the problem specification, for example by:

- (S1) enlarging the scope to a larger net $\Sigma' \supset \Sigma$ (adjoining additional observables or finer resolution), and/or
- (S2) enlarging the invariant package to $\mathcal{I}' \supset \mathcal{I}$ (declaring additional invariants as primitive).

We refer to either operation as a *scope/invariant-package enlargement*.

5.2 Scope-level closure lemma

Proposition 5.7 (Closure by exhaustion (scope-level)). *Fix an admissibility datum $(\Sigma, \mathcal{I}, \sim)$. If a demand $\mathcal{D}$ is not primitive under $\sim$, then $\mathcal{D}$ is not determined by the invariant package $\mathcal{I}$ at scope Σ : there exist realizations $X \sim Y$ with $\mathcal{D}(X) \neq \mathcal{D}(Y)$. In particular, imposing $\mathcal{D}$ as part of the problem specification requires a scope/invariant-package enlargement in the sense of Definition 5.6.*

Proof. If $\mathcal{D}$ is not primitive, then by Definition 5.5 there exist $X \sim Y$ with $\mathcal{D}(X) \neq \mathcal{D}(Y)$. Since X and Y agree on $\mathcal{I}$ at scope Σ , $\mathcal{D}$ is not fixed by $\mathcal{I}$ on Σ . Therefore $\mathcal{D}$ cannot be demanded as a same-scope consequence of $\mathcal{I}$ without enlarging the primitive data, i.e. enlarging scope and/or invariant package. $\square$

Remark 5.8 (Further closure principles). In separate companion notes one may strengthen Proposition 5.7 into statements that exclude additional same-scope “admissibility gates” once a declared invariant package is fixed. The present paper uses only the elementary scope-level statement above.

6 Conditional completion theorem under admissibility

We now state the bridge completion claim.

Theorem 6.1 (Conditional completion under admissibility). *Fix $\ell > 0$ and scope $\Sigma = \Sigma_\ell$. Assume:*

(H1) (Route 1 + Paper 4) the macroscopic invariant package (M1)–(M4) holds on Σ_ℓ [1, 4].

(H2) (Admissibility) the admissibility datum treats scope Σ_ℓ and invariant package $\mathcal{I} = \{(M1)–(M4)\}$ as primitive for the mass-gap statement.

Then the mass-gap statement is closed under this admissibility: no additional same-scope requirement is needed to assert a Hamiltonian spectral gap in the vacuum representation generated by $\mathcal{A}_\ell$.

Moreover, any further demand that is not determined by (M1)–(M4) (for example, stable-manifold RG structure or sharp $t \rightarrow 0$ local fields) constitutes a scope enlargement by Proposition 5.7.

Proof. Assumption (H1) asserts the macroscopic invariant package (M1)–(M4), in particular the Hamiltonian spectral gap (M4) for the OS reconstruction in the vacuum representation generated by $\mathcal{A}_\ell$. Assumption (H2) declares that, at scope Σ_ℓ , the mass-gap statement treats precisely the invariant package (M1)–(M4) as primitive. Therefore no additional same-scope requirement can be imposed as part of the mass-gap statement beyond (M1)–(M4). The final sentence follows from Proposition 5.7. $\square$

Remark 6.2 (What remains external). Theorem 6.1 is conditional in two independent ways: the Route 1 macroscopic invariant package depends on explicit UV identification assumptions (Paper 4), and the closure statement depends on the chosen admissibility datum. Neither conditionality is hidden: both are part of the scope bookkeeping. See Table 1 for a status map of the principal nontrivial inputs.

7 What is eliminated at macroscopic scope

We record a list of demands that are non-primitive at macroscopic scope unless explicitly built into the scope.

Proposition 7.1 (Examples of non-primitive upgrades at macroscopic scope). *In an admissibility datum with scope Σ_ℓ and invariant package (M1)–(M4), the following additional requirements are non-primitive unless explicitly built into the scope/invariant package:*

*(E1) **Global stable-manifold RG structure:** global manifold/trajectory structures used to organize RG control beyond what is required to witness macroscopic clustering.*

*(E2) **Renormalization flow parameterization:** any requirement that an RG “time” parameterization be treated as primitive input for spectral extraction, beyond OS semigroup time.*

*(E3) **Interpretive narratives about mixing:** any requirement that a particular interpretive narrative (e.g. information-theoretic) be treated as a primitive mechanism; the proofs use divergence/mixing quantities only as quantitative contraction witnesses.*

*(E4) **Sharp-field ultraviolet regularity:** pointlike operator-valued distributions obtained as $\ell \downarrow 0$ (or $t \rightarrow 0$) limits.*

Remark 7.2. Proposition 7.1 is not polemic. It does not deny that these items may be useful analytic tools or desirable refinements. It asserts only that they do not factor through (M1)–(M4) and therefore cannot be demanded as primitive at macroscopic scope without enlarging the scope.

8 Sharp local fields as a non-primitive demand

This section restates the scope enlargement of Paper 5 in the admissibility language.

Definition 8.1 (Sharp-field scope enlargement). The *sharp-field scope* enlarges Σ_ℓ by adjoining an ultraviolet limit $\ell \downarrow 0$ (equivalently $t \rightarrow 0$ in gradient-flow language) that produces pointlike operator-valued distributions.

Remark 8.2 (Why this is not forced by macroscopic invariants). At fixed $\ell > 0$, (M1)–(M4) are statements about a bounded net and its OS reconstruction. Recovering pointlike fields typically requires additional ultraviolet regularity inputs (energy bounds, nuclearity/phase-space conditions, scaling limits), cf. Haag [11] and Buchholz–Wichmann [12], and see Paper 5 [5] for a companion discussion at the level of bounded nets. These are not determined by (M1)–(M4), hence the demand is non-primitive by Proposition 5.7.

9 Conclusion and open upgrades

Route 1 provides a macroscopic mass-gap construction on a bounded local net at fixed $\ell > 0$, with OS positivity and an explicit Hamiltonian spectral gap, conditional on explicit UV identification inputs. Papers 2–5 isolate minimal proof primitives (finite-depth entrance, OS semigroup extraction) and clarify that sharp local fields are a strict ultraviolet scope enlargement.

The present capstone adds an admissibility bookkeeping layer. Under an admissibility datum that treats the macroscopic scope Σ_ℓ and invariants (M1)–(M4) as primitive, Route 1 together with the UV package closes the corresponding mass-gap statement. Optional upgrades within enlarged scopes include: deriving the UV identification assumptions from first principles; constructing sharper fields and broader temperedness; and upgrading the spectral statement toward additional relativistic structure.

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